

Flutter Developer Hiring Task

Task Overview

Task goal is a mobile-first skin analysis companion designed to help users understand the current state of their skin. A user uploads or captures a photo of their face, an AI model scans it, marks the areas where skin issues are detected, and returns a clear, easy-to-understand skin analysis.

Most skincare apps stop at generic tips. This task focuses on showing the user what is actually happening on their skin — visually marked, region by region — and explaining it in plain language.

The goal of this task is to build a functional mobile prototype that demonstrates the core product experience: secure sign-in, a smooth onboarding flow, and an AI-powered face scan that produces an annotated result image along with a skin analysis report.

This is not about building a production-ready application. This is about demonstrating:

- Product thinking
- Mobile development skills
- UI / UX design
- Backend architecture
- AI integration
- Execution ability

What You Need to Build

A functional mobile application prototype for the task built around three pillars:

1. Authentication
2. Onboarding
3. AI Face Scan & Skin Analysis (the core feature)

The app should feel personal, clean, and trustworthy. The objective is an experience a user would genuinely return to whenever they want to check in on their skin.

Scope Rules (Important)

- Focus on core product functionality.
- Features may be simplified, but they must work end-to-end.
- UI / UX quality will play a major role in evaluation.
- Authentication and onboarding must be built using Firebase.
- Backend and data storage may use Firebase or Supabase.
- The AI face scan must use an open-source AI model.
- A focused, working MVP beats an unfinished large product.
- You are free to introduce additional features if they create meaningful value.

Application Flow

1. Intro / Landing Experience

When users open the application they should see a calm, premium welcome — for example:

"Understand your skin, clearly."

Followed by short onboarding screens introducing the product:

- Meet your personal skin companion.
- Scan your face in just a few seconds.
- See exactly where skin issues show up.
- Get a simple skin analysis you can actually act on.

Note: All example copy and screens below are reference templates only, not fixed requirements. Make thoughtful product decisions that best fit the overall vision and user experience.

2. Authentication (Firebase or superbase — Required)



Users should be able to:

- Sign up using Email
- Login using Email
- Continue with Google

- Continue with Apple

Requirement: Authentication must be implemented using Firebase Authentication. Keep it simple and frictionless.

3. User Onboarding

Collect the information the app needs to personalise the experience.

Personal Information

- Name
- Age
- Gender

Skin Information

- Skin type (oily, dry, combination, normal, sensitive)
- Primary skin concerns (acne, dark spots, dullness, wrinkles, redness, etc.)
- Known sensitivities or allergies (optional)

Goals

- Clearer skin
- Reduce acne
- Even out skin tone
- Reduce signs of ageing
- Improve hydration
- Build a consistent routine

Notification Preferences (optional)

- Scan reminders
- Routine reminders
- Progress updates

Note: These fields are not hard requirements. Adapt, simplify, or expand them based on your product thinking, as long as the app stays aligned with the core objective.

4. AI Face Scan & Skin Analysis (Core Feature)

This is the most important part of the task. Prioritise it.

Step 1 — Capture / Upload

The user can capture a selfie inside the app or upload an existing photo of their face. Provide light guidance for a good scan (front-facing, good lighting, no heavy filters). A basic image-quality check is a plus.

Step 2 — AI Analysis (Open-Source Model — Required)

The uploaded face must be processed by an open-source AI model. The model should detect facial regions and identify skin issues. You are free to combine a face-detection / landmark model with a skin-condition detection or segmentation approach.

Issues to detect (examples — detect as many as you reasonably can):

- Acne / breakouts
- Dark spots / pigmentation
- Dark circles
- Redness / irritation
- Wrinkles / fine lines
- Enlarged pores
- Oiliness
- Dryness
- Uneven texture
- Dullness

Step 3 — Annotated Result Image (Required)

Generate a result image of the user's face with the problem areas clearly marked — using overlays, highlighted regions, bounding markers, or a heat-style map. A user should be able to glance at the image and immediately see where the issues are on their face.

Step 4 — Skin Analysis Report

Alongside the image, present a clear, readable report:

- Detected issues, with location and a rough severity (e.g. mild / moderate / high)
- An overall skin score or summary
- A short, plain-language explanation for each issue

- Simple suggestions or next steps (general guidance only)

Keep the experience focused on awareness and clarity. This is a skin-awareness product, not a medical diagnosis tool — frame results accordingly.

5. Results & History

Users should be able to view their past scans and revisit previous results.

Bonus: Let users compare scans over time and track progress on the concerns they care about.

6. Profile & Settings

Users should be able to manage:

- Profile — personal information, skin profile, and goals
- Notification settings
- Privacy settings — including the ability to delete their photos and scan data
- Account preferences

Handling Face Photos (Important)

Because the app handles photos of people's faces, store and handle them responsibly. At a minimum: keep each image tied to the authenticated user, never expose images publicly, and give users a clear way to delete their own data.

Design & UX Guidelines

The product should feel:

- Personal
- Trustworthy
- Clean
- Premium
- Modern
- Reassuring

The product should never feel:

- Clinical

- Judgmental
- Overwhelming
- Cluttered

Focus heavily on:

- Simplicity
- Visual hierarchy
- A clear scan → result flow
- Smooth navigation
- A result screen that is genuinely satisfying to look at

UI / UX quality will play a significant role in judging.

Tech Expectations

Mobile Application (Required): React Native + Expo

Authentication & Onboarding (Required): Firebase or Supabase

Backend & Data: Node.js

Database: Firestore, Supabase Postgres, or similar

AI (Required): An open-source AI model. You are free to choose the tools and approach that best support the skin analysis.

Example building blocks:

- Face detection / landmarks — e.g. MediaPipe Face Mesh, RetinaFace, MTCNN, dlib
- Skin-condition detection / segmentation — open-source models, including ones available through Hugging Face

You may run inference however you prefer — on-device, your own API, or a hosted open-source endpoint — as long as the underlying model is open-source.

Points to Note

- UI / UX quality is extremely important.
- The AI face scan and annotated result image are the core deliverable — prioritise them.
- Authentication and onboarding must use Firebase.

- Core flows must work end-to-end.
- A focused MVP is better than an unfinished large product.
- You are free to use AI tools to increase productivity. What matters is how smartly you use them and how much ownership you show.
- Product thinking and execution quality matter more than feature count.

Submission Instructions

Required Submission

1. Demo Video (3–5 minutes) — walk through the full product: authentication → onboarding → face scan → annotated result → skin analysis.
2. Functional Mobile Application — submit the APK file.

iOS Testing Process

If your submission qualifies for final evaluation, our team may provide TestFlight access for deployment and testing. Participants with their own Apple Developer Account may share a TestFlight build directly. Additional tester emails can be provided if needed.

Source Code: Source code submission is strictly not required.

How to Submit: All submissions are made through the Google Form below.

Google Form: <https://forms.gle/wVVzibYezTDd9Gmu8>

Deadline: Thursday 2nd End of the day

The Opportunity

The person who completes this task by the deadline and delivers the best, most functional and user-friendly app will receive a full-time offer at Humanity Founders as a founding-level member, with a high CTC and base pay.

Closing Note

This task is more than an assignment — it is a chance to work on one of the most exciting product ideas out there and to shape your own future while doing it.

It is a test of skills over resumes. In the era of AI you are completely free to use any AI tool to do this — what matters is how smartly you use it and how much ownership you bring.

We want to work with people who enjoy challenges: the ones who keep pushing even when things break, and don't stop until it finally works. Once you're onboard, you'll be working with a team of genuinely talented people on seriously exciting stuff.

Thank you for being here, and all the very best. We're looking forward to connecting with the person who gets this across the line.